Şeyma Karadereli

Ph.D. in Mathematics

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Research Interests

Low-dimensional topology, contact topology, and singularity theory. Main focus on real algebraic realization of contact structures and Milnor open books supporting contact structures.

Education

Ph.D. in Mathematics, Boğaziçi University 2020 - Aug 2026

Advisor: Ferit Öztürk
Dissertation Title: Real Algebraic Overtwisted Contact Structures on Milnor Fillable 3-Manifolds

M.S. in Mathematics, Boğaziçi University 2017 - 2020

Advisor: Ferit Öztürk
Thesis Title: Algebraic Overtwisted Contact Structures on the 3-Sphere

B.S. in Mathematics, Boğaziçi University 2012 - 2017

Academic Positions

Graduate Teaching Assistant Jan 2020 - Present

Boğaziçi University

Visiting Ph.D. Student Feb 2023 - Jun 2023

Erwin Schrödinger International Institute for Mathematics and Physics

Graduate Teaching Assistant Mar 2019 - Jan 2020

Özyeğin University

Publications and Preprints

Real Algebraic Overtwisted Contact Structures on 3-Spheres (with F. Öztürk), *Algebraic & Geometric Topology* **25** *(2025)*, **DOI: 10.2140/agt.2025.25.1377**

Real Milnor Open Books and Algebraic Overtwisted Contact Structures on Lens Spaces (with F. Öztürk), *(In preparation)*

Talks

Real Algebraic Overtwisted Contact Structures on 3-Spheres and Lens Spaces,

AWM Special Session on Recent Developments in Symplectic and Contact Geometry and Topology *Joint Mathematics Meeting, Chicago* Jan 2027

Real Algebraic Overtwisted Contact Structures on 3-Spheres,

Young Women in Geometric Topology *Hausdorff Research Institute for Mathematics, Bonn* Jul 2026
IMBM Istanbul Topology Days *Boğaziçi University, Istanbul* May 2026
6th Austrian Day of Women in Mathematics *University of Graz* Feb 2026
Winter School in Low-Dimensional Topology and Singularity Theory *Erdős Center, Budapest* Jan 2023

Honors, Scholarships, Awards

National Scholarship Program for PhD Students, *TUBITAK (The Scientific and Technological Research Council of Türkiye)* 2020-2024

Junior Research Fellowship, *Erwin Schrödinger International Institute for Mathematics and Physics* 2023

National Scholarship Program for MSc Students, *TUBITAK* 2017-2019

Undergraduate Scholarship Program, *TUBITAK* 2014-2017

Undergraduate Scholarship Program, *Government of Turkiye* 2012-2014

Teaching Experience

Teaching Assistant Coordinator, Boğaziçi University 2022

Department of Mathematics, team of approx. 10 graduate TAs

- Distributed tasks among TAs
- Organized exam proctoring; confirmed exam dates, determined external proctor needs and communicated with the Dean's Office
- Ran orientation for incoming teaching assistants
- Served as the main point of contact between TAs and the department

Graduate Teaching Assistant, Boğaziçi University Jan 2020 - Present

Conducted weekly problem sessions, selected exercises together with the instructor and prepared written solutions to problem sets.
Held office hours, graded exams, homework and quizzes.
Courses:

- Math 101 - Calculus 1 (4 semesters)
- Math 102 - Calculus 2 (3 semesters)
- Math 201 - Matrix Theory (6 semesters)
- Math 105 - Calculus for Social Sciences (3 semesters)
- Math 111 - Introduction to Mathematical Structures (3 semesters)
- Math 162 - Discrete Mathematics (2 semesters)
- Math 131 - Calculus for a Single Variable
- Math 132 - Calculus of Several Variables
- Math 221 - Linear Algebra
- Math 231 - Advanced Calculus I (3 semesters)
- Math 325 - Matrix Groups
- Math 327 - Number Theory
- Math 471 - Topology

Graduate Teaching Assistant, Özyeğin University Mar 2019 - Jan 2020

Weekly office hours, exam gradings for the courses:

- Calculus I-II for Engineering Students
- Calculus for Social Sciences
- Discrete Mathematics

Outreach, Mentoring, and Service

Local Organizer, *IMBM Istanbul Topology Days, Boğaziçi University* 2026

Organizer, *Writing Group, Boğaziçi University* 2026
Regular sessions where faculty members, graduate students meet to work on their own writing projects.

Volunteer Tutor, *Refugee children in Türkiye* 2021-2023
Helping refugee children with mathematics lessons.

Mentor, *Directed Reading Program Türkiye* 2021
Mentored an undergraduate student in a reading project on classification of surfaces using Morse Theory.

Accessibility Tutor, *Boğaziçi University (referred by GETEM)* 2018
Provided semester long one on one instruction to a student with dyslexia taking Math 202 - Differential Equations in the Department of Mathematics.

Organizing Team, *IMAGINARY Mathematics Exhibition, Istanbul* 2020
Coordinated exhibition guides; the exhibition was cancelled in its first days due to COVID-19.

Workshops and Conferences Attended

31st Gökova Geometry-Topology Conference May 2026
Gökova Geometry Topology Institute *Muğla, Turkiye*

IMBM Istanbul Topology Days May 2026
Boğaziçi University *Istanbul, Turkiye*
A Conference on Developments in Contact Topology, Mapping Class Groups, and Beyond

Mapping Class Groups and Beyond May 2025
Middle East Technical University *Ankara, Turkiye*
A workshop to celebrate contributions of Mustafa Korkmaz

The Semester on Singularities and Low-Dimensional Topology Spring 2023
Erdős Center *Budapest, Hungary*
Winter School in Low-Dimensional Topology and Singularity Theory, The Singular Workshop, The Low-Dimensional Workshop.

Frontiers in Geometry and Topology Summer School and Conference Aug 2022
ICTP, The Abdus Salam International Centre for Theoretical Physics *Trieste, Italy*

Geometry-Topology Student Workshop on Khovanov Homology May 2022
Gökova Geometry Topology Institute *Muğla, Turkiye*

Braids in Low-Dimensional Topology Apr 2022
ICERM *Virtual Attendee*

Tech Topology Summer School Jul 2021
Georgia Institute of Technology *Remote*

Mini School on Singularities of Surfaces Nov 2019
Galatasaray University *Istanbul, Turkiye*

Bosphorus Real Geometry Conference Oct 2019
IMBM, Istanbul Center for Mathematical Sciences *Istanbul, Turkiye*

26th Gökova Geometry-Topology Conference Jun 2019
Gökova Geometry Topology Institute *Muğla, Turkiye*

Contact Topology Workshop Mar 2019
Hacettepe University *Ankara, Turkiye*

Topology and Geometry in Low Degrees, Workshop Dec 2018
IMBM, Istanbul Center for Mathematical Sciences *Istanbul, Turkiye*

25th Gökova Geometry-Topology Conference Jun 2018

Gökova Geometry Topology Institute *Muğla, Turkiye*

References

Prof. Ferit Öztürk (Ph.D. advisor)
Department of Mathematics
Boğaziçi University, Istanbul, Türkiye
ferit.ozturk@bogazici.edu.tr

Prof. Mohan Bhupal
Department of Mathematics
Middle East Technical University, Ankara, Türkiye
bhupal@metu.edu.tr

Assoc. Prof. Bahar Acu
Department of Mathematics
Pitzer College, Claremont, CA , US
bahar.acu@pitzer.edu

Prof. Dr. Özlem Beyarslan (Teaching reference)
Department of Mathematics
Boğaziçi University, Istanbul, Türkiye
ozlem.beyarslan@bogazici.edu.tr

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